

RESEARCH PAPER

The Impact of 6G Network Performance on Manufacturing Efficiency in Smart Factories

A Diagnostic Analytics Study of Network Latency, Packet Loss, and Industrial Production Outcomes

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Abstract

As manufacturing transitions into Industry 4.0 and 5.0 paradigms, smart factories increasingly rely on ultra-low latency, high-reliability communication networks to synchronize autonomous machinery and AI-driven processes. While traditional manufacturing analytics focus heavily on mechanical health, this paper investigates the independent, critical role of telecommunications infrastructure — specifically 6G network performance — on production outcomes.

Utilizing a dataset of 100,000 operational records from Thales Group, advanced Key Performance Indicators (KPIs) were engineered to quantify the impact of network latency and packet loss on production speed and defect rates. The deployment of an interactive diagnostic dashboard reveals that network degradation can silently throttle efficiency and induce operational errors without triggering mechanical fault alarms, highlighting the necessity of a “connectivity-first” approach in modern industrial analytics.

1. Introduction

The integration of the Industrial Internet of Things (IIoT) and artificial intelligence within manufacturing environments demands continuous, real-time data exchange. In 6G-enabled smart factories, machines are highly interdependent; a delay in communication can cascade into significant production bottlenecks.

Historically, when manufacturing efficiency drops, organizations primarily investigate mechanical failures or process inefficiencies. However, a critical blind spot remains: the silent degradation of network performance. Stakeholders often lack clarity on whether a drop in production is due to a machine fault or micro-delays in network communication. This research addresses this gap by analyzing how latency variations and packet drops independently impact the defect rate, error rate, and overall manufacturing efficiency, ultimately preventing the misdiagnosis of production issues and the misallocation of network investments.

2. Methodology and Data Engineering

2.1 Dataset Overview

The analysis was conducted on an industrial dataset comprising 100,000 synchronized records of machine telemetry and network communication metrics. Variables included operational temperature, vibration, power consumption, network latency (ms), packet loss (%), quality control defect rates, and classified efficiency statuses (High, Medium, Low).

2.2 Feature Engineering and KPI Development

To isolate the impact of network stability from mechanical factors, novel metrics were engineered:

- **Network Quality Classification:** A dynamic banding system that categorizes network states into High (Stable), Medium, and Low (Unstable) based on median latency and packet loss thresholds.

- **Network Stability Index (NSI):** A weighted composite score (0 to 100) aggregating normalized latency and packet loss metrics to provide a single, top-level indicator of communication reliability.
- **Latency Sensitivity Score:** Evaluates the degree to which specific operation modes experience efficiency degradation per unit increase in network latency.

3. Key Analytical Insights

3.1 Network Degradation vs. Production Speed

The exploratory data analysis established a strong negative correlation between network latency and production speed. In highly automated states, machines waiting for AI-inferences or M2M (Machine-to-Machine) synchronization commands exhibited micro-stalls during high-latency periods. This results in a silent reduction in the units produced per hour, even when the machine's mechanical parameters (e.g., temperature, vibration) remain entirely within optimal operational thresholds.

3.2 Packet Loss and Defect Concentration

Packet loss proved to be a critical driver of operational errors and product defects. When packet loss exceeded baseline thresholds, the percentage of operational errors spiked proportionally. The diagnostic contours revealed specific “Defect Concentration Zones,” indicating that intermittent data drops cause asynchronous machine movements, directly translating to flawed manufacturing outputs that fail quality control checks.

3.3 Operation Mode Sensitivity

The analysis demonstrated that network sensitivity is not uniform across all machine states. Machines engaged in high-load or 'Active' operational modes exhibited a significantly lower tolerance for network instability compared to machines in 'Idle' or 'Standby' modes. This insight is critical for prioritizing network bandwidth allocation during peak production cycles.

4. Diagnostic Dashboard Architecture

To operationalize these findings, an enterprise-grade web application was developed using Streamlit and Plotly. The interactive framework empowers production managers and network engineers to visualize complex telecommunications data without requiring advanced data science expertise.

4.1 Core System Modules

Module	Description
Network Performance Overview	Real-time histograms and box plots tracking latency distributions and packet loss variations.
Network vs. Efficiency Dashboard	Interactive scatter plots and bar matrices correlating network quality bands with classified manufacturing efficiency.

Module	Description
Quality & Error Impact Panel	Regression modeling (OLS) visualizing the direct trajectory of error rates as a function of packet drops.
6G Optimization Insights	An automated diagnostic engine that calculates and flags critical threshold limits (Risk Zones) for latency and packet loss based on historical failures.

5. Strategic Recommendations and Conclusion

The findings from this research mandate a paradigm shift in how industrial analytics are approached. Network performance must be treated with the same level of diagnostic scrutiny as mechanical machine health.

Recommendations for Thales Group & Industry Partners

- **Implement 6G Network Slicing:** Prioritize guaranteed ultra-low latency network slices for machines operating in high-load modes to prevent asynchronous manufacturing errors.
- **Cross-Functional Diagnostics:** Integrate the Network Stability Index directly into the factory's primary Supervisory Control and Data Acquisition (SCADA) systems to prevent engineers from misdiagnosing network latency as mechanical faults.
- **Proactive Threshold Monitoring:** Establish automated alerts when packet loss enters the identified “Risk Zones” to pause production lines preemptively, thereby saving raw materials and minimizing defect rates.